Lesson 12.3 - Tests of Independence Using Chi-Square Distribution

> **Video:** Test of Independence Using Chi-Square Distribution by The Organic Chemistry Tutor **Chapter:** 12 - The Chi-Square Distribution

The big picture: are two categories related?

Lesson 12.1 compared **one** categorical variable with a distribution that H_0 stated in advance. Here there are **two** categorical variables measured on the same people, for example student type (freshman or senior) and daily study time (0-2, 2-4 or 4-6 hours). The question: **does knowing one tell you anything about the other?**

1. Count the combinations in a table. Each student falls into one row and one column. The result is a **contingency table** of counts, with row totals, column totals and a grand total.

2. What would "no relationship" look like? If study time had nothing to do with student type, freshmen and seniors would split across the study-time columns in the **same proportions**: the split of all 630 students. Overall, $223/630 = 35.4\%$ study 0-2 hours, so we'd expect 35.4% of the 310 freshmen to do so too:

$$E = 310 \times \frac{223}{630} = 109.7$$

3. Where $E = \dfrac{\text{row total} \times \text{column total}}{\text{grand total}}$ comes from. Use the multiplication rule for independent events (Lesson 5.5):

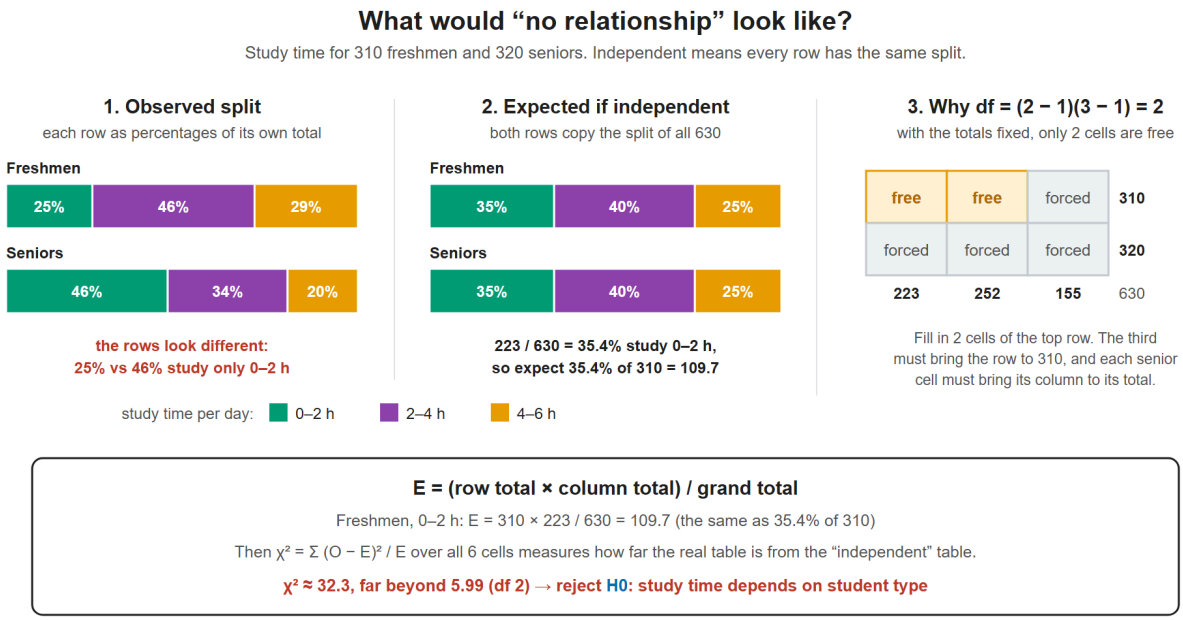
$$P(\text{row and column}) = P(\text{row}) \times P(\text{column}) = \frac{\text{row total}}{N} \times \frac{\text{column total}}{N}$$

To turn a probability into a count, multiply by N:

$$E = N \times \frac{\text{row total}}{N} \times \frac{\text{column total}}{N} = \frac{\text{row total} \times \text{column total}}{N}$$

So H_0 (independence) gives us every expected count, built from the table's own totals.

4. Then it's the same χ^2 as Lesson 12.1. Compare each observed count with its expected count, and add up $\frac{(O-E)^2}{E}$ over **every cell** of the table. A large χ^2 means the real table is far from the "independent" table.



Left: in the real data the freshman and senior rows look different. Middle: under independence both rows would have the same split. χ^2 measures the distance between these two pictures.

5. Why df $= (r-1)(c-1)$. The row and column totals are fixed. In each row, the last cell is forced by the row total, which leaves $c-1$ free cells per row. The last row is forced by the column totals, which leaves $r-1$ free rows. Here that is $(2-1)(3-1) = 2$ free cells.

6. Related, not "causes". Rejecting H_0 says the variables are **associated**. It doesn't say why. Seniors might study less for many reasons, and the test can't tell which.

How this compares with Lesson 12.1:

	Goodness of fit (12.1)	Independence (this lesson)
Variables	one	two
Expected counts come from	the proportions that H_0 states	the table's own row and column totals
df	$k-1$	$(r-1)(c-1)$
Test statistic	$\sum(O-E)^2/E$	$\sum(O-E)^2/E$ over all cells

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The idea

A **test of independence** checks whether two categorical variables in a **contingency table** are related.

	Formula
H_0	the variables are **independent**
H_a	the variables are **dependent** (related)
Expected count for each cell	$E = \dfrac{\text{row total} \times \text{column total}}{\text{grand total}}$
Test statistic	$\chi^2 = \sum \dfrac{(O-E)^2}{E}$ over **all** cells
Degrees of freedom	$(r-1)(c-1)$, where r = rows and c = columns

It is a **right-tailed** test: reject H_0 if χ^2 exceeds the critical value.

Example - study time vs type of student

Is the number of hours spent studying each day related to whether a student is a freshman or a senior? $\alpha = 0.05$.

Step 1 - complete the observed table

	0-2 h	2-4 h	4-6 h	**Row total**
Freshmen	76	143	91	**310**
Seniors	147	109	64	**320**
Column total	**223**	**252**	**155**	**630**

Step 2 - hypotheses and critical value

- H_0: study hours are **independent** of student type
- H_a: study hours **depend** on student type
- df $= (r-1)(c-1) = (2-1)(3-1) = 2$, right-tail area 0.05: $\chi^2_{\text{crit}} = 5.99$

Step 3 - expected counts

$$E = \frac{\text{row total} \times \text{column total}}{\text{grand total}}$$

Example: $E_{1,1} = \dfrac{310 \times 223}{630} \approx 109.7$. The other cells use their own row and column totals, for example $E_{2,3} = \dfrac{320 \times 155}{630} \approx 78.7$.

	0-2 h	2-4 h	4-6 h
Freshmen	109.7 (≈110)	124.0	76.3 (≈76)
Seniors	113.3 (≈113)	128.0	78.7 (≈79)

The rounded values still match every row and column total (e.g. 110 + 124 + 76 = 310).

Step 4 - test statistic (with the video's rounded expected counts)

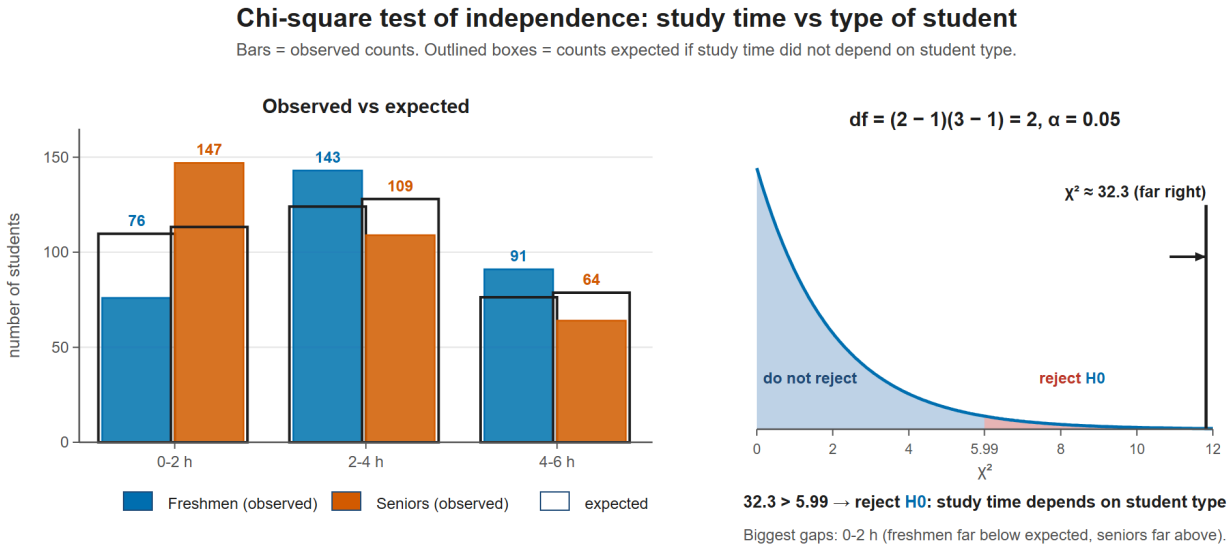
$$\chi^2 = \sum_{\text{all cells}} \frac{(O-E)^2}{E}, \qquad \text{for example, the first cell: } \frac{(76-110)^2}{110} = \frac{1156}{110} = 10.51$$

Cell	O	E	$(O-E)^2/E$
Freshmen, 0-2	76	110	10.51
Freshmen, 2-4	143	124	2.91
Freshmen, 4-6	91	76	2.96
Seniors, 0-2	147	113	10.23
Seniors, 2-4	109	128	2.82
Seniors, 4-6	64	79	2.85
Total			**≈ 32.3**

> **Note:** expected counts don't have to be whole numbers. Using the unrounded values (109.73, 124.0, 76.27, 113.27, 128.0, 78.73) gives $\chi^2 \approx 31.7$. The conclusion is the same.

Step 5 - decision

$32.3 > 5.99$, which is in the rejection region. **Reject H_0.** There is strong evidence that study time **depends** on student type. In the data, seniors are much more likely than freshmen to study only 0-2 hours a day.



Left: if study time were independent of student type, the bars would roughly fill the outlined expected boxes. The 0-2 hour group is far off for both freshmen and seniors. Right: $\chi^2 \approx 32.3$ is far beyond 5.99.

Key takeaways

- Under independence every row has the same split as the whole table. That is where expected count = (row total × column total) ÷ grand total comes from.
- $\chi^2 = \sum(O-E)^2/E$ over every cell, with df $= (r-1)(c-1)$.
- Right-tailed: a large χ^2 → reject independence → the variables are related.